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24 April 1957
SUPERSEDING
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MILITARY SPECIFICATION

VALVES, SHUTOFF, HYDROCARBON FUEL SERVICE 100 P.S.I. AND 150 P.S.I. WORKING PRESSURE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy and the Air Force.

1. SCOPE

1.1 Scope. This specification covers shutoff valves for gasoline service where absolute tightness after repeated operations is essential and the seat tightness is not made with a lubricant seal.

1.2 Classification. Valves shall be of the following classes, as specified (see 6.1):

Class 100 — 100 pounds per square inch (p.s.i.) maximum working pressure.

Class 150 — 150 p.s.i. maximum working pressure.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

PPP–B–601 — Boxes, Wood, Cleated Plywood.

PPP–B–621 — Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL–P–3 — Packaging and Packing for Delivery of Valves, Fittings, and Flanges (Except for Oxygen Systems).

JAN–P–100 — Packaging and Packing for Overseas Shipment: General Specification.

MIL–P–116 — Preservation, Methods of.

MIL–C–132 — Crates, Open Wood, Maximum Capacity 2,500 Pounds.

MIL–B–233 — Boxes, Repair Parts.

MIL–B–857 — Bolts, Nuts, Studs, and Tap Rivets (and Material for Same).

MIL–D–963 — Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).

FSC 4820

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MIL-N-994 — Naval Brass Bars, Plates, Rods, Sheets, Strips, Wire, Forgings, and Structural and Special Shaped Sections.

MIL-P-5315 — Packing, "O" Ring, Hydrocarbon Fuel Resistant.

MIL-G-6183 — Gaskets and Sheet Gasket Material: Synthetic-Rubber and Cork Composition.

MIL-R-6855 — Rubber; Synthetic, Sheet, Molded, and Extruded, for Aircraft Applications.

MIL-P-15024 — Plates, Identification — Information and Marking for Indentification of Electrical, Electronic, and Mechanical Equipment.

MIL-T-15071 — Technical Manuals for Mechanical and Electrical Equipment.

MIL-B-16541 — Bronze, Valve: Castings.

MIL-F-17292 — Format for List of Repair (Spare) Parts for Shipboard Mechanical and Electrical Equipment.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 601 — Rubber; Sampling and Testing.

MILITARY

MIL-STD-10 — Surface Roughness, Waviness, and Lay.

MIL-STD-129 — Marking for Shipments and Storage.

MIL-STD-130 — Identification Marking of U. S. Military Property.

DRAWINGS

BUREAU OF SHIPS

B-104 — Fittings, Pipe, Flanged, Composition.

B-176 — Flanges, Composition, Silver Brazing.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 **Qualification.** Valves furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.2).

3.2 **Material.** Except as otherwise specified hereinafter, all materials shall be nonsparking and suitable for use with high-octane gasoline, kerosene, jet fuel, and short-period contact with sea water. Material shall be such as to withstand temperatures from minus 40° to plus 212° Fahrenheit (F.). Materials shall be capable of withstanding steam at 212°F. for periods up to 12 hours. The material of the parts shall be as follows:

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3.2.1 *Bodies.* The bodies shall be of bronze conforming to Specification MIL–B–16541.

3.2.2 *Valve handwheels.* The valve handwheels shall be of cast bronze construction.

3.2.3 *Bolts or studs.* The bolts or studs shall be of Naval brass conforming to Specification MIL–N–994, and the form and threading shall be in accordance with Specification MIL–B–857.

3.2.4 *Rubber materials.* Rubber materials used in seat or disc construction shall be in accordance with class 1, grade 60 of Specification MIL–R–6855 except that a durometer hardness of 60 minus 5 plus 10 (55–70) will be permitted. The adhesion of the rubber to the metal core shall be a minimum of 15 pounds per inch when tested in accordance with method 8031 of Test Method Std. No. 601. The durometer hardness test shall be conducted on the finished seating surface of the coated plugs or discs and shall be noted as the instantaneous reading. Between five and ten readings shall be averaged for this purpose. The requirements for the cold bend test of Specification MIL–R–6855 shall be waived. The maximum compression set requirement shall be 50 percent in lieu of 40 percent.

3.2.5 *Gaskets and washers.* Gaskets and washers shall be adequately designed of materials suitable for the service intended. Nonmetallic gaskets shall conform to class 1, grade firm, of Specification MIL–G–6183. The "O" ring gaskets shall conform to Specification MIL–P–5315.

3.3 Construction and design.

3.3.1 *Construction.* The valves shall be of a construction which will withstand repeated operations with a minimum of wear on working parts for working pressures up to 100 p.s.i. for class 100 valves, and up to 150 p.s.i. for class 150 valves.

3.3.2 *Design.* Valves shall be of a design having low-pressure drop and full pipe area at any point throughout the fluid passage in the body. Valves shall have round ports and shall be designed so as to eliminate pockets where gasoline may be trapped when the valve is installed in any position. The valve shall be designed to resist damage by high-impact shock. Such shock shall not change the position in which the valve has been set. For plug type valves which do not reseat in the open position there shall be no more than 0.050 inch radial clearance between the plug or disc and the body when the valve is in the open position.

3.3.2.1 Valves shall be equipped with a suitable indicator to indicate the position of the valve plug or disc.

3.3.2.2 Valves shall be quick and easy operating and shall not require more than 2¼ turns of the handwheel for operation. Operating parts of the mechanism shall be of such combinations of materials that will have adequate strength and resistance to galling or wear when operated without lubrication. However, such parts shall be lightly lubricated to prevent corrosion while in storage. Where the operating mechanism is separate from the fluid handling portion of the valve, provision for lubrication of the mechanism will be permitted if means are provided to prevent contamination of the fluid handled.

3.3.2.3 The finish for mating surfaces of flanges and flange fittings shall be a serrated or phonographic finish with 60 to 80 cuts per inch of face width 0.002-inch to 0.003-inch-deep cuts.

3.3.3 *Inlet and outlet connection.* Inlet and outlet connections of the valve shall be integral with the body. The connections shall be flanged. Flanged connections for class 100 valves shall conform to the dimensions shown on Drawing B–104 with the drilling in accordance with Drawing B–176. Class 150 valve flanged connections shall conform to the dimensions shown on Drawing B–104 with an additional ⅛-inch thickness; the drilling shall be as shown on Drawing B–176.

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3.4 Size. The size of valves shall conform to the face to face dimensions shown in table I for the size specified (see 6.1).

TABLE I. *Face to face dimensions for classes 100 and 150 valves.*

Size	Face to face
Inches	*Inches*
1	6⅝
1¼	6⅝
1½	8
2	9¼
2½	10⅜
3	11½
3½	12¼
4	12¾
6	16⅝
8	22
10	24¼
12	32
14	34

3.5 Weights. Weights shall not exceed those specified in table II for the size specified (see 6.1).

TABLE II. *Weight.*

Size	Weights, class 100 (maximum)	Weights, class 150 (maximum)
Inches	*Pounds*	*Pounds*
1	[1]	[1]
1¼	[1]	[1]
1½	75	80
2	85	90
2½	100	105
3	[1]	[1]
3½	[1]	[1]
4	200	210
6	385	400
8	[1]	[1]
10	1,175	1,225
12	[1]	[1]
14	[1]	[1]

[1] To be as approved by the bureau or agency concerned.

3.6 Seat tightness and valve tightness. The valves shall be capable of withstanding the air freon test for seat tightness specified in 4.3.1.1 and the air freon test for valve tightness specified in 4.3.1.2 without showing any leakage or other signs of weakness.

3.7 Hydrostatic pressure. The valves shall be capable of withstanding the hydrostatic pressures shown in table III without any leakage, sweating, or visible deformation (see 4.3.2).

TABLE III. *Hydrostatic pressure.*

	Class 100	Class 150
(a) With valve in open position; for strength and porosity.	200 p.s.i	300 p.s.i.
(b) With valve in closed position for seat tightness. Pressure to be applied alternately in each direction.	150 p.s.i.	225 p.s.i.

3.8 Cycling. The valves shall be capable of withstanding the cycling test of 4.3.3 without any difficulty of operation, leakage or signs of excessive wear.

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.1).

3.10 Marking. The valves shall be provided with an identification plate conforming to type C of Specification MIL–P–15024. The identification plate shall be made of brass material. The identification data appearing on the identification plate shall conform to Standard MIL–STD–130.

3.10.1 *Additional marking.* Coated plugs or discs shall have the month and year, in which the plug or disc is coated, stamped, engraved, or burned on a part of the plug or disc other than the seating surface, where it can be readily noted. Such marking shall not affect the strength or durability of the rubber coating. In addition to marking of the plug or disc, a tag with the same date (month and year) shall be wired to the valve and handwheel.

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3.11 Drawings.

3.11.1 *Preliminary drawings.* Each bidder shall submit preliminary drawings with his bid to adequately describe the equipment he proposes to furnish. If a bidder proposes to furnish equipment which previously has been supplied and for which the bureau or agency concerned has approved drawings on record, reference to the approved drawings will be acceptable in lieu of submitting preliminary drawings.

3.11.2 *Working drawings.* Completely detailed working drawings in accordance with Specification MIL–D–963 shall be submitted to the bureau or agency concerned by the contractor for completion of contract unless otherwise specified (see 6.1).

3.12 Technical manuals. Technical manuals shall be furnished in accordance with type B of Specification MIL–T–15071; as specified (see 6.1).

3.13 Workmanship. Valves shall be free from defects that affect their appearance or their operation. Castings shall be clean, sound and free from blow holes, hard spots, porosity, cracks and other injurious defects. They shall be smooth and well cleaned both inside and outside, and all fins and roughness shall be removed. Castings shall not be repaired, plugged, impregnated, brazed or burned. Surface finishes where indicated by the symbol $\surd$ shall be in accordance with Standard MIL–STD–10. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests at a Government laboratory. Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.4.

4.1.1 Prior to authorization of qualification tests, a drawing of the proposed design showing design and materials shall be submitted to the Bureau of Ships for preliminary design approval.

4.2 Examination (at place of manufacture).

4.2.1 *Visual and dimensional examination.* Each valve shall be visually and dimensionally examined by the inspector to verify compliance with this specification, the drawings, and with the valve granted qualification. Any valve failing to comply shall be rejected.

4.3 Test procedure (at the place of manufacture).

4.3.1 The inspector shall witness the various tests performed on each valve for acceptance. Each valve shall be tested as specified in 4.3.1.1 through 4.3.2.

4.3.1.1 *Air freon test for seat tightness.* One flange shall be blanked off and the plug or disc turned to the closed position. Freon at 5 p.s.i. shall be introduced into the valve and the pressure built up to 75 p.s.i. by the further introduction of air or nitrogen. The tightness of the valve seats shall be checked with a halide torch. The pressure shall be reduced to 5 to 10 p.s.i. and the valve seats checked again for tightness. This procedure shall be repeated in the reverse direction, the other flange being blanked off. It is recommended that this test follow 4.3.1.2.

4.3.1.2 *Air freon for valve tightness.* Both flanges shall be blanked off with the plug or disc turned to the open position. Freon and air or nitrogen shall be introduced into the valve as specified in 4.3.1.1 to a pressure of 75 p.s.i. The tightness of the stem, bonnet, and any other joints and the porosity of the body shall be checked with a halide torch.

4.3.1.3 Any leakage or other signs of weakness during the tests specified in 4.3.1.1 or 4.3.1.2 shall be cause for rejection.

4.3.2 *Hydrostatic tests.* Each valve shall be subjected to the hydrostatic tests with water to determine conformance with 3.7. Any leak-

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age, sweating, or visible deformation under these tests shall be cause for rejection.

4.3.3 *Cycling test.* One valve of each size manufactured under a contract or order (other than sizes which have previously passed qualification tests at a Government laboratory) shall be subjected to a minimum of 7,000 cycles of operation with kerosene. Class 100 valves shall be operated at 100 p.s.i. during cycling; class 150 valves at 150 p.s.i. Any difficulty of operation, leakage, or signs of excessive wear during or at the completion of this test shall be cause for rejection. The contractor will not be required to so test any sizes of valves of identical construction to those which have successfully passed such tests under a previous contract or order.

4.4 Qualification tests. The manufacturer shall submit one 6-inch valve and any other size valves deemed necessary by the Bureau of Ships to the Government laboratory for qualification. The materials and design of the valve offered for qualification shall conform to this specification. Qualification testing shall be performed at the Government laboratory as follows:

4.4.1 *Cycling tests.* Valves offered for qualification tests shall be subjected to a minimum of 7,000 cycles of operation with high-octane gasoline. Class 100 valves shall be operated at pressures of 25, 50, 75, and 100 p.s.i. during the cycling. In addition to these pressures, class 150 valves shall also be subjected to operation at 150 p.s.i. Before and after the cycling, valves shall be subjected to the following tests:

4.4.1.1 *Hydrostatic tests.* Hydrostatic tests shall be performed as specified in 4.3.2.

4.4.1.2 *High-octane gasoline.* High-octane gasoline pressure shall be applied alternately on both sides of the plug or disc to determine seat tightness, tightness of body and bonnet joints, stem packing and any other joints by use of a fume-detecting apparatus. Class 100 valves shall be subjected to 125 p.s.i. and class 150 valves shall be subjected to 200 p.s.i.

4.4.1.3 *Air freon tests.* Air freon tests shall be performed as specified in 4.3.1.1 and 4.3.1.2, and the provisions of 4.3.1.3 shall apply.

4.4.2 *Steam test.* The valves shall be subjected to steam at 212°F. for a period of at least 12 hours. After cooling, the valves shall be given a cycling test for 500 cycles as specified in 4.4.1. The valves shall then be tested as specified in 4.4.1.1, 4.4.1.2, and 4.4.1.3.

4.4.3 Any leakage during the tests specified in 4.4.2 will be cause for disqualification.

4.4.4 Ease and reliability of operation shall be observed and reported on by the testing activity.

4.4.5 *Shipboard service evaluation.* In addition to the foregoing laboratory qualification tests, a 6 months service test aboard a naval vessel will be required. Satisfactory service of valves so installed is necessary prior to final qualification approval. The number and sizes of valves for shipboard installation will be determined by the Bureau of Ships.

4.4.6 *Shock tests.* Valves shall be shock tested to determine conformance with 3.3.2.

5. PREPARATION FOR DELIVERY

5.1 Cleaning, preservation, and packaging.

5.1.1 *Level A.*

5.1.1.1 The valves shall be cleaned, preserved, and port openings closed as specified in Specification MIL–P–3. The interior of valves shall be preserved by spraying or flushing with preservative type P–7, P–8, or P–9 in accordance with Specification MIL–P–116.

5.1.1.2 *Maintenance parts.* Repair parts (when specified) shall be cleaned, preserved

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and packaged in accordance with Specification MIL–P–116. Items generally shall be packaged one item per unit package unless used in sets or in quantities greater than one.

5.1.2 *Level C.* The cleaning, preservation, and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.* When specified (see 6.1), each valve cleaned, preserved and packaged in accordance with level A or C, as specified (see 6.1) shall be unit packed in a cleated plywood or nailed wood box conforming to Specification PPP–B–601, or style 2, 2½, or 3 of Specification PPP–B–621, or in a nailed wood unsheathed crate conforming to Specification MIL–C–132.

5.2.2 *Anchoring and waterproofing.* Anchoring, strapping, and waterproofing of contents of unsheathed crates shall be in accordance with the appendix of Specification MIL–C–132. Blocking, bracing, and cushioning shall be in accordance with Specification JAN–P–100, except that the use of shredded newspaper or excelsior will not be permitted.

5.2.3 *Level B.* Unless otherwise specified in the contract or order, each valve cleaned, preserved and packaged in accordance with level A or C, as specified (see 6.1), shall be unit packed in a nailed wood box conforming to style 2, 2½, or 3 of Specification PPP–B–621, or in a nailed wood unsheathed crate conforming to Specification MIL–C–132.

5.2.4 *Level C.* Equipment, cleaned, preserved and packaged in accordance with level A or C, as specified (see 6.1) shall be packed in a manner to insure safe delivery and acceptance at destination. Containers or method of shipment shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.3 Repair parts.

5.3.1 *Packing.* Where onboard repair parts are destined for a ship that is not fitted with bin- or drawer-type stowage (see 6.1), the onboard repair parts shall be packed in type M or W boxes conforming to Specification MIL–B–233 and shall be overpacked in shipping containers conforming to Specification MIL–C–132 or PPP–B–621. The gross weight shall not exceed 200 pounds. Where onboard repair parts and tools are destined for a ship that is fitted with bin- or drawer-type stowage (see 6.1), both onboard and stock repair parts shall be packed in nailed wood or cleated plywood boxes as specified in 5.2.1 or 5.2.3.

5.3.2 *Index list of repair parts.* An index list of repair parts shall be inserted in each shipping container containing onboard repair parts. The list shall conform to Specification MIL–F–17292.

5.4 Technical publications.

5.4.1 Technical manuals, pamphlets, and handbooks, as applicable, shall be packaged in accordance with method 1C–3 of Specification MIL–P–116. When specified (see 6.1), one or two copies of the technical manuals, shall be packed in the shipping container holding the unit. Bulk quantities of the technical manual shall be packed in cleated plywood or nailed wood boxes as specified in 5.2.1 or 5.2.3.

5.5 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Class required (see 1.2).

(c) Size and weight (see tables I and II).

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(d) Repair parts, if required (see 3.6).

(e) If detailed working drawings are not required (see 3.8.2).

(f) Number of technical manuals required (see 3.9).

(g) Selection of applicable levels of cleaning, preservation and packaging and packing (see section 5).

(h) Whether valves are to be unit packed (see 5.2.1).

(i) Type of stowage for onboard repair parts, if required (see 5.3.1).

6.2 **Qualification.** With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL–16468, whether or not such products have actually been so listed by that date.

6.2.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D.C.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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